Lokesh Kanna Rajaram

rajaramlokeshkanna@gmail.com | Portfolio | LinkedIn | +1(716) 520 7064| GitHub

Summary

Data Scientist with experience on building scalable analytics and **ML pipelines** across **computer vision**, **NLP**, and **time-series data**. Proven ability to translate large, noisy datasets into actionable insights through **statistical modeling**, distributed data processing, and production-style experimentation. Experienced working with **Python**, **SQL**, **Spark**, and cloud-based data systems to support data-driven decision-making.

TECHNICAL SKILLS

Programming Languages & Databases: Python, R, MySQL, Pytorch, Hadoop, Apache Spark, Kafka, Pandas, MongoDB, Informatica, AWS.
Tools & Platforms: PowerBI (DAX), Tableau, Docker, GitHub/Git, Microsoft Office, Generative AI, LLM, Amazon Web Service (AWS), Redshift, ETL Pipelines, CI/CD Pipelines, Statistical Analysis, Cloud Infrastructure, Data Visualization.
Certifications: Informatica Cloud (IICS), Cloud Practitioner – CLF-C02.

EDUCATION

University at Buffalo, The State University of New York, NY, USA — **Dec 2025**
Master of Science, Data Science

WORK EXPERIENCE

Data Analyst – City of Houston, Houston, Texas — **Mar 2026 - Present**

- Engineered an **automated ingestion pipeline in Python** utilizing **REST APIs** and **secure authentication keys, accelerating raw data acquisition by 40% to** programmatically download and centralize complex inspection datasets from Sewer AI for analysis.
- **Automated ETL pipelines within AWS Redshift** to process and structure complex, multi-source telemetry data across 6,200 miles of infrastructure, **reducing upstream data preprocessing times for AI research workflows by 30%**.
- **Enriched multi-modal dataset integrity** by programmatically aligning real-time SewerAI text/video metadata with ArcGIS Pro spatial maps, **increasing data asset reliability to 98%**.

Research Scientist Volunteer – University at Buffalo (SMILE lab), Buffalo, NY — **Dec 2025 - Present**

- Built **GAN, DNN** model to **denoise** and **enhance ultrasound image data, improving data usability for modelling** tasks.
- **Built quantitative evaluation frameworks** to measure noise, artifacts, and domain shift, identifying **25-40% clutter reduction** and **15-20% generalization drops** across probes and acquisition settings.
- Optimized inference **pipelines for real-time clinical workflows, achieving 50ms latency** to validate feasibility for live, point-of-care image enhancement.

Computer Vision Data Scientist Capstone - Nissha Medical Technologies, Buffalo, New York — **Aug 2025 - Dec 2025**

- Designed and deployed a **real-time computer vision pipeline** using **YOLOv8, OpenCV,** and **PyTorch** to detect manufacturing defects under strict latency constraints, reducing unplanned **machine downtime by 12%.**
- Engineered a **hybrid ML + Statistical Process** Control monitoring system, combining **deep learning-based object detection with rolling-window SPC analytics,** early failure detection and **reducing defect rates from 4% to 3%.**
- Built a production-style **analytics workflow** using **Pandas** and **SciPy** to track **temporal quality drift (dimensional and colour metrics, ΔE)**, **transforming raw model outputs** into **actionable predictive maintenance signals**.

PROJECTS

Student Lease Break Advisor (GenAI + RAG System) Link — **Dec 2025**

- Built a production-grade **RAG-based legal assistant** using **Gemini, LangChain, FAISS,** and **Streamlit** that analyzes lease agreements against state tenant laws, reducing unsupported **LLM** responses by **~70% through dual vector-store separation**.
- Engineered a rate-limit–aware embedding pipeline with batched **semantic indexing** and throttling, enabling stable document ingestion under **Gemini** free-tier constraints and **eliminating 429 API errors across 100+ document chunks**.
- Designed legally constrained prompting and retrieval logic that enforces citation-backed outputs and explicit refusal behaviour, achieving deterministic**, low-hallucination responses.**

Amazon Book Review Using Big Data Pipeline Link — **May 2025**

- Built an **end-to-end big data analytics pipeline** using **Hadoop and PySpark** to **ingest, clean, and transform 1M+ Amazon book reviews**, enabling **large-scale feature engineering** and model-ready datasets.
- **Conducted scalable experimentation and evaluation**, **validating model performance across distributed datasets** and translating results into actionable insights on sentiment trends and review quality.
- **Engineered NLP features at scale** using **Spark ML (Tokenizer, StopWordsRemover, HashingTF, IDF)**, **improving model training efficiency by 40%** while achieving **90.4% classification accuracy**.